

Week 3: Statistical Analysis and Hypothesis Testing in Python

Project Overview

This project performs statistical analysis and hypothesis testing on a **Student Performance Dataset** using Python.

The main objective is to determine whether there is a statistically significant difference in **Math Scores** between students who **completed test preparation** and students who **did not complete test preparation**.

Objective

To apply statistical methods in Python and understand whether test preparation is associated with a difference in students' Math Scores.

Research Question

Is there a significant difference in the average Math Scores of students who completed test preparation and those who did not?

Hypothesis

Null Hypothesis (H_0)

There is **no significant difference** in the average Math Scores between students who completed test preparation and those who did not.

Alternative Hypothesis (H_1)

There is a **significant difference** in the average Math Scores between students who completed test preparation and those who did not.

Significance Level

The significance level used for the test is:

$\alpha = 0.05$

Dataset

The project uses the file:

```
Original_data_with_more_rows.csv
```

Important variables used in the analysis:

- **TestPrep** – indicates whether the student completed test preparation.
- **MathScore** – student's Math Score.

The dataset contains **30,641 observations**.

Statistical Test

An **Independent Samples t-test (Welch's t-test)** is used because the analysis compares the mean Math Scores of two independent groups.

The two groups are:

1. Students who completed test preparation
2. Students who did not complete test preparation

Welch's t-test is used without assuming that the two groups have equal variances.

Technologies Used

- Python
 - Pandas
 - NumPy
 - SciPy
 - Matplotlib
 - Jupyter Notebook / Python Script
 - GitHub
-

Project Structure

```
Week_3_GitHub_Project/  
|  
├── Dataset.csv/  
|   └── Original_data_with_more_rows.csv  
|  
├── graphs/  
|   ├── math_score_histogram.png  
|   └── math_score_boxplot.png
```

```
|  
|— Analysis.ipynb/  
|   |— hypothesis_testing.py  
|  
|— report/  
|   |— Week_3_Hypothesis_Testing_Report.docx  
|  
|— README.md
```

Python Analysis

The Python program performs the following steps:

1. Imports the required Python libraries.
2. Loads the student dataset.
3. Cleans the `MathScore` and `TestPrep` variables.
4. Separates students into two test-preparation groups.
5. Calculates the sample size, mean, and standard deviation.
6. Performs an independent samples Welch's t-test.
7. Calculates the difference between group means.
8. Calculates a 95% confidence interval.
9. Creates a histogram of Math Scores.
10. Creates a boxplot comparing the two groups.
11. Makes the final hypothesis-testing decision.

Visualizations

Histogram

The histogram shows the distribution of Math Scores for students in the two test-preparation groups.

Boxplot

The boxplot compares the Math Score distributions between students who completed test preparation and those who did not.

Results

The Welch independent samples t-test produced:

- **t-statistic:** approximately `-25.51`
- **p-value:** approximately `$1.89 \times 10^{-141}$`
- **95% Confidence Interval:** approximately `[-4.95, -4.25]`

Since the p-value is much smaller than **0.05**, the null hypothesis is rejected.

Conclusion

There is a **statistically significant difference** in Math Scores between students who completed test preparation and those who did not.

Therefore, the analysis provides strong statistical evidence that the two test-preparation groups have different average Math Scores.

How to Run the Project

1. Clone the repository

```
git clone YOUR_GITHUB_REPOSITORY_URL
```

2. Open the project folder

```
cd Week_3_GitHub_Project
```

3. Install the required libraries

```
pip install -r requirements.txt
```

4. Run the Python program

```
python notebooks/hypothesis_testing.py
```

The program will perform the statistical analysis and display the graphs.

Learning Outcomes

Through this project, we learn how to:

- Work with real-world datasets using Python.
- Perform data cleaning.
- Calculate descriptive statistics.
- Formulate null and alternative hypotheses.
- Perform an independent samples t-test.
- Interpret p-values.
- Calculate and interpret confidence intervals.
- Create statistical visualizations.
- Make data-based conclusions.
- Document statistical analysis in a GitHub repository.

Project

Week 3 Task – Statistical Analysis and Hypothesis Testing in Python

Topic: Student Performance and Test Preparation

Analysis: Independent Samples t-test